

Title: Java Basics & OOP Fundamentals

1. Program 1: Basics (Loops & Conditions)

Code:

```
public class BasicsDemo {  
    public static void main(String[] args) {  
  
        int[] marks = {85, 60, 90, 45, 70};  
  
        System.out.println("Student Results:");  
  
        for (int i = 0; i < marks.length; i++) {  
  
            if (marks[i] >= 50) {  
                System.out.println("Student " + (i + 1) + " Pass");  
            } else {  
                System.out.println("Student " + (i + 1) + " Fail");  
            }  
        }  
    }  
}
```

Output:

```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>notepad BasicsDemo.java

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>javac BasicsDemo.java

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java BasicsDemo
Student Results:
Student 1 Pass
Student 2 Pass
Student 3 Pass
Student 4 Fail
Student 5 Pass

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>|
```

2. Program 2:

Encapsulation

Code:

```
class Student {
    private String name;
    private int marks;

    public Student(String name, int marks) {
        this.name = name;
        this.marks = marks;
    }

    public String getName() {
        return name;
    }

    public void setMarks(int marks) {
        this.marks = marks;
    }

    public void display() {
        System.out.println("Name: " + name + ", Marks: " + marks);
    }
}
```

```

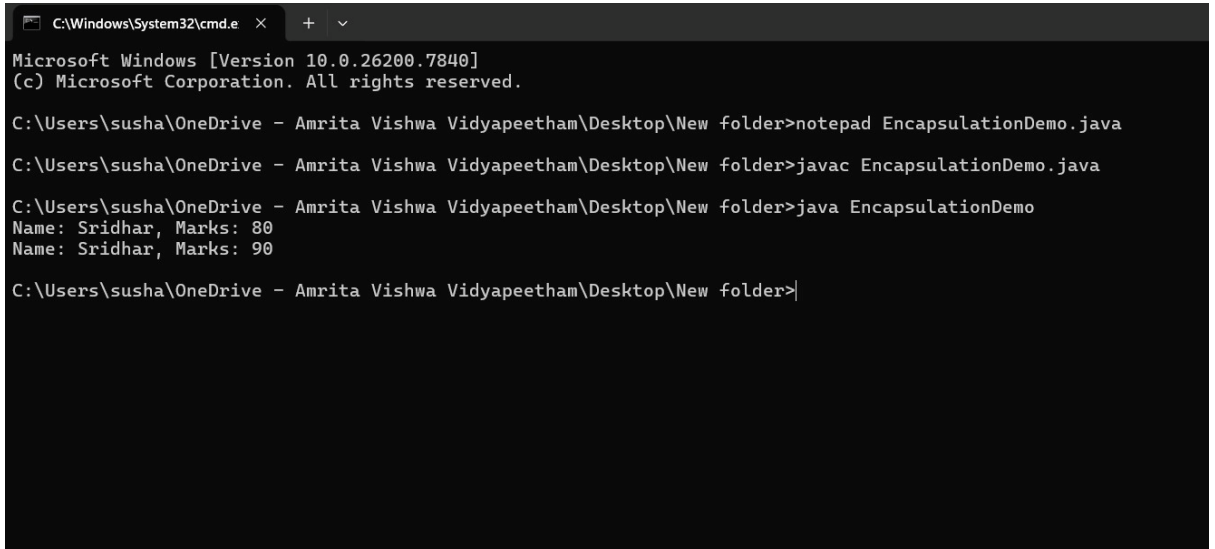
public class EncapsulationDemo {
    public static void main(String[] args) {

        Student s1 = new Student("Sridhar", 80);
        s1.display();

        s1.setMarks(90);
        s1.display();
    }
}

```

Output:



```

C:\Windows\System32\cmd.e  ×  +  ▾
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>notepad EncapsulationDemo.java

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>javac EncapsulationDemo.java

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java EncapsulationDemo
Name: Sridhar, Marks: 80
Name: Sridhar, Marks: 90

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```

3. Program 3:

Inheritance & Polymorphism

Code:

```

class Animal {
    public void sound() {
        System.out.println("Animal makes sound");
    }
}

class Dog extends Animal {
    public void sound() {
        System.out.println("Dog barks");
    }
}

class Cat extends Animal {

```

```
    public void sound() {  
        System.out.println("Cat meows");  
    }  
}  
  
public class InheritanceDemo {  
    public static void main(String[] args) {  
  
        Animal a;  
  
        a = new Dog();  
        a.sound();  
  
        a = new Cat();  
        a.sound();  
    }  
}
```

Output:

```
C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java InheritanceDemo  
Dog barks  
Cat meows  
  
C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>|
```

Encapsulation : Data hiding using private variables and methods

Inheritance: One class acquires properties of another class

Polymorphism: Same method behaves differently for different objects

Abstraction: Hides internal implementation and shows functionality

These programs demonstrate fundamental OOP concepts used in real-world applications such as student management and system design.